

Restaurant Recommendation System

Introduction

In today's world, there are many restaurant options available, which makes it difficult for users to decide where to eat. This project helps users by recommending restaurants based on their preferences using machine learning techniques. It improves user experience by providing quick and relevant suggestions.

Objective

- To build a smart recommendation system
 - To help users find similar restaurants
 - To reduce time in decision-making
 - To use machine learning for better suggestions
 - To create a simple and user-friendly web interface
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Problem Statement

Users face confusion when choosing restaurants due to too many options and lack of proper guidance. Traditional searching takes time and may not give the best results. This system solves the problem by giving personalized recommendations based on restaurant data and reviews.

Technologies Used

- **Python** – for backend logic
- **Flask** – for building web application

- **Pandas** – for data processing and cleaning
 - **Scikit-learn** – for machine learning model
 - **HTML & CSS** – for designing frontend
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Dataset

The project uses the **Zomato dataset**, which contains:

- Restaurant name
- Ratings
- Cuisines
- Reviews
- Cost for two people
- Location

This data helps the system understand restaurant features and user preferences.

Data Preprocessing

- Removed missing (null) values
- Removed duplicate records
- Cleaned rating values (converted to float)
- Converted cost column into numeric format
- Converted text to lowercase
- Removed punctuation from reviews

These steps improve the accuracy of the model

Model Used

- **TF-IDF Vectorization**
Converts text data (reviews) into numerical form
- **Cosine Similarity**
Measures similarity between restaurants based on reviews

This combination helps in finding restaurants with similar features.

Working of System

1. User enters a restaurant name
 2. System searches the dataset
 3. Finds similar restaurants using cosine similarity
 4. Filters results (optionally by location)
 5. Displays top recommended restaurants
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Web Application

The project includes 3 main pages:

- **Home Page**
Shows project overview and navigation
- **Recommendation Page**
User enters restaurant name
- **Result Page**
Displays recommended restaurants in table format

Output

The system shows:

- Restaurant Name
- Rating
- Cuisines
- Cost for two people

This helps users compare and choose easily.

Features

- Simple and clean user interface
- Fast recommendation system
- Uses real-world dataset
- Shows multiple restaurant details
- Easy navigation between pages